

A decorative background in the top half of the cover, featuring a dense arrangement of triangles in various shades of pink, red, teal, and white, creating a geometric pattern.

User Guide

Scottish Longitudinal Education Outcomes (LEO) – University extract dataset

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Glossary

CIS	Customer Information System; DWP population spine used for LEO linkage
DEA	Digital Economy Act 2017; governs access to data through the Researcher Accreditation Service
DfE	Department for Education (UK)
DWP	Department for Work and Pensions (UK)
HESA	Higher Education Statistics Agency
HMRC	His Majesty's Revenue and Customs
IDBR	Inter-Departmental Business Register; source of employer SIC codes
JUVOS	DWP longitudinal benefits data system
LEO	Longitudinal Education Outcomes
NINO	National Insurance Number
ONS	Office for National Statistics
PAYE	Pay As You Earn; HMRC payroll reporting system
RTI	Real Time Information System; HMRC payroll and employment data
SA	Self-Assessment; HMRC tax data related to self-employed or additional income
SIC	Standard Industrial Classification; industry codes
SRS	Secure Research Service; ONS Trusted Research Environment
SRSA / SRSA 2007	Statistics and Registration Act 2007
UKSA	UK Statistics Authority; responsible for researcher accreditation

1. Introduction and Context

Purpose and Audience

This User Guide is primarily designed to support accredited researchers, analysts and other data users who are considering using this dataset for public good research. It provides essential information about the dataset, including its structure, scope and research potential, to help users develop robust and feasible research applications.

This guide is intended to be used alongside the corresponding Data Dictionary, which details the variables and technical specifications of the dataset. Together, these documents aim to ensure transparency, usability and accessibility of the data.

Dataset Overview

The **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** contains information on students from Scottish Higher Education Institutions derived from the Higher Education Statistics Agency (HESA) student data. These records are linked to information on employment, earnings and benefits held by His Majesty's Revenue and Customs (HMRC) and the Department for Work and Pensions (DWP). Additional variables on the standard industrial classification of employment are sourced from the Inter-Departmental Business Register (IDBR).

The dataset is constructed from pseudonymised data received by the Scottish Government following the LEO data linkage process (see Section 4: *Data Linkage Methodology* for details). It is already a linked dataset and does not contain variables that enable further linkage to datasets outside the LEO process. The following data sources are already linked through this dataset:

- Higher Education Statistics Agency (HESA) data on students at UK publicly funded higher education institutions and some alternative providers (university data only).
- Employment data from HMRC's Real Time Information System (RTI), which contains information formerly collected through P45 (employment) and P14 (PAYE) forms.

- Income data from HMRC Self-Assessment tax returns.
- Benefit receipt and labour market information from DWP systems, including the National Benefit Database, Labour Market System and JUVOS, along with geographic region data.
- Standard Industrial Classification (SIC) codes derived from the Inter-Departmental Business Register (IDBR).

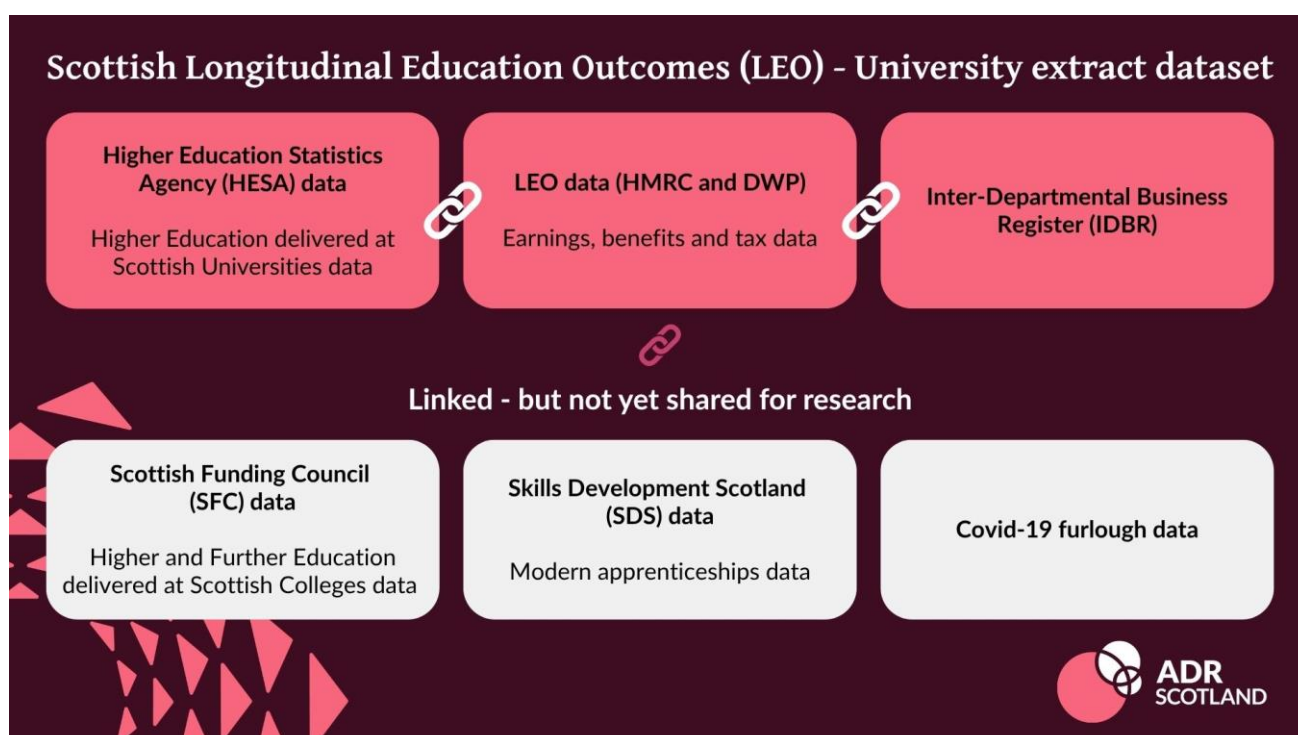


Figure 1. Linked data sources within the Scottish Longitudinal Education Outcomes (LEO) - University extract dataset

Motivation for Linkage

The LEO data linkage project was initiated to meet the UK Government's [2015 manifesto](#) that contained commitments to:

- Publish more earnings and destination data for Further Education courses, and
- Make more information openly available to potential students so that they can make decisions informed by the career paths of past graduates.

The Scottish Government [joined the LEO project in 2017](#), with the aim of publishing earnings and destinations data for learners completing post-16 education in Scotland.

The LEO dataset supports a range of Government objectives:

- **Improving information for students and parents.** LEO enables reporting of graduate salaries once individuals are established in their careers. Linked data also increases the coverage and quality of information available to students through public sources such as Discover Uni: [Discover Uni Home | Discover Uni](#).
- **Improving the quality of information on labour market outcomes.** LEO supports better alignment between courses, student aspirations and labour market needs by providing more detailed evidence on expected outcomes by background, subject, qualification and institution.
- **Enhancing labour market productivity and economic performance.** More robust evaluations of the effectiveness of education and training programmes help education providers tailor learning to labour market requirements and improve returns for students and employers.
- **Strengthening value-for-money assessments and policy evaluation.** LEO supports medium and long-term analysis of earnings returns, qualification pathways and economic impacts to inform policy design, including cost-benefit analysis across interventions, courses and qualifications.
- **Improving government efficiency.** Linked administrative data provides more comprehensive and timely information than survey-based approaches alone. For example, LEO complements and enhances Scotland's coverage in the Graduate Outcomes Survey, which is conducted 15 months after graduation. More information about the Graduate Outcomes Survey can be found here: [Thousands of graduate voices. One survey - Graduate Outcomes](#)
- **Providing secure access for research.** LEO offers a low-cost alternative to other forms of data collection and evaluation, allowing added value to be derived from existing data. The secure access to LEO through ONS SRS allows for analysis of outcomes within a secure research environment while maintaining data confidentiality.

Research Potential and Policy Relevance

This dataset provides substantial potential for research that can directly inform national priorities across education, skills and wellbeing. By linking higher education records with detailed employment, earnings and benefits data, it enables robust analysis of graduate pathways, regional labour market outcomes and inequalities affecting specific groups. The dataset supports a wide range of policy frameworks, including **GIRFEC**, **The Promise**, **Bringing Hope, Building Futures** and the **Scottish Attainment Challenge**.

The structure of the dataset has similarities to the established LEO model used in England (noting that Scottish LEO does not yet include schools data), which has underpinned significant research on attainment, mobility and post-16 trajectories. As Scotland's LEO resource develops, it will offer growing opportunities for cross-sector and longitudinal analysis aligned with ADR Scotland's Strategic Research Themes, strengthening the evidence base on skills, outcomes and inequality across the life course.

Further detail is provided in Section 5.

Comparable Datasets

The Department for Education (DfE) holds an extensive LEO dataset covering individuals in education in England. [DfE LEO Extract](#)

The Graduate Outcomes Survey, conducted by HESA four times a year, surveys graduates approximately 15 months after completing their studies. It provides a snapshot of post-graduation education and employment outcomes, whereas LEO provides multi-year outcomes based on linked tax records. [Thousands of graduate voices. One survey - Graduate Outcomes](#)

2. About the Data

Data Owner

Scottish Ministers

Contact Details

Contact details for queries about the dataset:

Scottish Government

Education Analytical Services

Advanced Learning and Skills Analysis

Statistics team: FHEstatistics@gov.scot

Data Collection Details

The HESA data contains information on Higher Education (HE). In Scotland, this only relates to HE delivered at Higher Education Institutions (HEIs), and not any HE delivered through Scottish Colleges. HESA collects data from higher education institutions through the annual HESA Student Return, which includes information about students and their courses.

Institutions have a statutory obligation to submit this data.

The employment and earnings data has been derived from HMRC administrative IT systems which, as with any large-scale recording system, are subject to possible errors with data entry and processing. The core purpose of this process is to collect tax from those who are eligible to pay it through this mechanism, which results in incomplete coverage of earnings data. Employers were previously not required to supply information to HMRC for individuals who earn below the tax threshold, although for large employers, these individuals are thought to be included due to common methods of data transfer to HMRC. HMRC data also provides the main two sources of input to the Inter-Departmental Business Register (IDBR) - which itself is the source of SIC codes of employers included in the LEO data.

The benefits data included in the LEO standard is taken from DWP administration systems. Due to the size and technical complexity of these systems, cuts of the data are taken at regular intervals. These are used to create longitudinal datasets, with start and end dates of benefit claims and the type of benefit.

Geography region data is accurate from 2012/13 when the current DWP system went live (October 2012). It is possible that address information is accurate before 2012/13, but only if individuals did not move from this address.

All data used in this process to construct the LEO data is drawn from administrative sources, and consequently takes time to process and collate.

Latest Release Date

The most recent linkage of the **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** was completed in April 2025, with updates undertaken annually as part of a scheduled linkage round each spring. The next planned update is due to take place in April 2026, with no anticipated delays or changes to the current annual linkage round.

Data Scope and Coverage

The dataset includes earnings, employment and benefits information alongside education information for UK-domiciled students who have studied at a Scottish Higher Education Institution.

Information on earnings includes amounts earned per tax year for both Pay As You Earn (PAYE) and Self-Assessed (SA) earnings. Information on employment includes flags for sustained employment, industry of employment and geographic location during employment per tax year.

Information on benefits includes flags to denote whether the individual was in receipt of active labour market benefits per tax year. Information on education includes characteristics

of the individual (sex, age, etc.) along with information about their course (level of study, subject, etc.).

Time Period Covered

Data is defined over two separate time periods. There is a time period for the education data linked to the tax data, which currently spans from the 2003/04 academic year to the 2019/20 academic year, and a time period for the tax/benefits data linked to the education data, which currently spans from the 2005/06 tax year to the 2021/22 tax year.

Geographical Coverage

The data covers UK-domiciled students from all Higher Education Institutions in Scotland, and their earnings and employment anywhere within the UK. Overseas earnings and employment are not captured, as these are not available through HMRC. Geographic location during employment is available at local authority level.

3. Data Quality

Understanding the quality of this dataset is essential for conducting robust and reliable analysis. This section outlines how the data is processed, linked and de-identified, and provides information on its comparability across years and regions. It highlights the strengths and known limitations of the dataset, including any issues with completeness, consistency or changes in methodology. By reviewing this section, you will gain insight into the reliability of the data and how best to interpret it in your research.

Data processing

The dataset is created by linking education, employment, earnings and benefits data. Prior to linkage, each data provider applies its quality assurance procedures on data to ensure it is suitable for linkage.

Once linked, data cleaning procedures are carried out on the data to ensure its suitable for analysis. The main cleaning procedures focus on employment and earnings data.

Employment spells are checked for invalid or implausible dates, merged where overlapping, and any missing employment start and end dates have a minimum and maximum possible date established. These minimum/maximum periods are narrowed down using surrounding employment records and out-of-work benefits records and then imputed using a distribution of known employment gaps.

Earnings data is cleaned by removing errors, duplicates and unmatched negative values, and linked to the employment data to ensure earnings have a valid accompanying employment spell.

Cleaned earnings and employment data is linked to education data using a pseudonymised ID – and derived variables are created using –such as sustained employment flags to indicate which tax years graduates were in sustained employment.

Additional variables on geography, industry and benefits are also linked in from their respective datasets, which all undergo basic cleaning procedures, mainly concerning missing values and duplicates.

The LEO linkage is reviewed annually – allowing for new data to be incorporated and processes updated as required.

For full detail, see the Scottish LEO Methodology Document (see Document Links further below).

Data Linking and De-Identification

The Scottish Government (SG) share education data with the Department for Education (DfE), who collate education administrative data for linkage purposes. DfE compile a dataset from education administrative data (including data from Wales and Scotland) containing personal identifiers – National Insurance (NI) numbers, names, dates of birth, gender and postcode – as well as a pseudo-identifier. DfE retain a mapping between this pseudo-identifier and any unique IDs.

DfE transfer the dataset to the Department for Work and Pensions (DWP), who hold personal identifiers on their Customer Information System (CIS). DWP match their CIS dataset to the education dataset before performing data linkage on DWP and HM Revenue and Customs (HMRC) tax and benefits records.

The linked datasets, which retain no personal identifiers post-linkage, are then transferred to DfE, the Welsh Government and the Scottish Government.

The Scottish Government receive linked data with pseudo-identifiers that allow for education data (also containing the same pseudo-identifiers) to be matched to the linked dataset, thereby creating a Longitudinal Education Outcomes (LEO) dataset.

Data Comparability

The data is comparable across regions and years but will be subject to real-world changes and differences (i.e. those in employment in London will likely earn more than those employed outside London, and earnings in later years will naturally be higher than earnings in earlier years due to wage inflation). Methodology and definitions are comparable across years, but tax data becomes cleaner over time, meaning that employment patterns are more likely to be accurate in more recent years. The methodology document outlines the data cleaning steps performed on the employment data.

Synthetic Data Availability

Synthetic data is currently not available for this dataset.

Document Links

Please find below links to key documentation and supporting materials for the dataset. Some documents require access to secure platforms.

- **Data Dictionary:** Available through ONS Metadata Works (login required):
<https://ons.metadata.works/browser/dataset/1861122/0/2091935>
- **Metadata Record:** Descriptive metadata for the Scottish LEO – University extract dataset is also available through ONS Metadata Works.
<https://ons.metadata.works/browser/dataset/1861122/0?tab=descriptiveMetadata>
- **Methodology Document:** The LEO methodology documents will be published by Scottish Government in Summer 2026 and will be accessible via:
<https://www.gov.scot/collections/colleges-universities-young-people-training-and-employment-statistics/>. If access to the methodology documentation is required before publication, please contact FHEstatistics@gov.scot.

Providing Feedback on the Data

Researchers are encouraged to report issues, inconsistencies or suggestions for improvement.

For any feedback on the dataset, please contact:

Scottish Government

Education Analytical Services

Advanced Learning and Skills Analysis

Statistics team: FHEstatistics@gov.scot

Community Groups

Although Scotland does not currently have its own LEO-specific research community, Scottish researchers can engage with the established LEO Research Community led by the Department for Education (DfE) in England. This community provides access to resources, published research and learning opportunities, including those developed through the ADR England Research Community Catalysts, which support cross-sector collaboration and capacity building around LEO data. Further information and links to join the community can be found on the ADR UK website: <https://www.adruk.org/data-access/flagship-datasets/longitudinal-education-outcomes/>.

As Scotland's LEO programme grows, establishing a dedicated Scotland-focused community network may become a future option. To register your interest, please contact: ADRScotland@gov.scot

4. Data Linkage Methodology

Linkage Overview

This data linkage brings together education records and labour market information under a well-defined legal and governance framework that permits government departments to share personal data for research, evaluation and the assessment of education, training and employment policy.

The LEO linkage is delivered by DWP using the Customer Information System (CIS) as the population spine. Education data from DfE is matched to CIS to generate a pseudonymised identifier, which then enables linkage to employment, earnings and benefits data.

Match rates are consistently high (around 95%), although they vary by data source and learner characteristics and some mismatches or missed matches are inevitable due to data quality.

Overall, the linkage enables the construction of LEO datasets that support a broad range of analytical work contributing directly to key government priorities.

Legal and Governance

1. Legal Gateway for Data Sharing

The data linkage is enabled by several statutory gateways that permit government departments to share and use personal data for research, evaluation and assessment of education, training and employment policy:

Education and Skills Act 2008 (Sections 87–88)

Allows information to be used or disclosed for “assessment functions” of UK, Welsh and Scottish Ministers or devolved authorities, defined as:

- Evaluating the effectiveness of training or education (s. 87(4)(a));
- Assessing policy in relation to the provision of training or education; and (s. 87(4)(b))

- Assessing policy in relation to social security or employment as it affects the provision of or participation in training or education (s. 87(4)(c))

Only specific categories of information may be shared, defined as information that:

- Is about an individual and is held by the Secretary of State for the purposes of any function relating to Social Security (s.87(2));
- Is about an individual and is held by the Secretary of State or a devolved authority and which relates to any training or course of education undertaken by the individual (s.87(3)); or
- relates to income tax or tax credits to a person for use in connection with the exercise of an assessment function of the Secretary of State or a devolved authority (s.88(1))

Apprenticeships, Skills, Children and Learning Act 2009 (Section 122)

Permits information sharing about individuals who are former teachers, to another Secretary of State to enable or facilitate the exercise of any of the Secretary of State's functions relating to education or training.

Small Business, Enterprise and Employment Act 2015 (Section 78)

Amended the Education and Skills Act 2008 to expand the gateway (at Sections 87 and 88 of that Act) which now enables tax-related data held by HMRC and social security data held by DWP to be shared and linked to school pupil information, further education and higher education student information.

2. Lawful Basis for Processing

The processing of personal and special category data is justified under:

UK GDPR

- processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller; (Article 6(1)(e))

Data Protection Act 2018

- Is necessary for archiving purposes, scientific or historical research purposes or statistical purposes, (Schedule 1. Part 1.4(a))
- is carried out in accordance with Article 89(1) of the UK GDPR (as supplemented by section 19), and (Schedule 1. Part 1.4(b))
- is in the public interest. (Schedule 1. Part 1.4(c))

Each Data Controller remains responsible for ensuring their own compliance with data protection legislation, including the oversight of any processors or third-party providers.

3. Governance Processes

Data Protection and Research Safeguards

Processing must comply with Article 89(1) UK GDPR and Section 19 DPA 2018, requiring safeguards such as data minimisation, pseudonymisation and restrictions on re-identification.

Re-identification controls:

- DWP manages pseudonymisation.
- Only DWP can authorise re-identification by any other party (DPA 2018, Part 6, s.171)

Restrictions on Individual Rights

Under DPA 2018 Schedule 2, Part 6, Paragraph 27, certain UK GDPR rights (e.g., access, rectification, objection) may be disapplied if exercising them would seriously impair research or statistical purposes, provided:

- Article 89 safeguards are in place
- Results are not published in a form that identifies individuals.

Human Rights Considerations

Data sharing engages Article 8 ECHR (right to private and family life), but the interference is justified because:

- It is lawful, necessary and proportionate
- It supports economic wellbeing
- Data is not used to make decisions about identifiable individuals.

4. Restrictions on Use

- Data must be used only for public-interest research, evaluation and statistical purposes related to education, training, employment and related policy.
- Shared data cannot be used to produce measures or decisions affecting specific individuals.
- Publication must occur only in anonymised, non-disclosive form.

- The agreement governing the data sharing is not legally binding and does not provide indemnity from legal action; each party remains accountable for compliance.

Linkage Provider and Spine

The **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** is constructed by matching education data held by DfE to the Customer Information System (CIS), which is used by DWP. This then allows a successful link to be matched to the labour market outcomes of matched individuals.

CIS contains a record for all individuals that have been issued with a National Insurance Number (NINO), including children whose parents have made a child benefit claim relating to them. CIS is DWP's primary source of customer information and the population spine for this linkage. DWP carry out all aspects of the linkage, linking education data to DWP matching variables and creating a pseudonymised ID that links matched individuals to their labour market outcomes. Only pseudonymised data is sent back to the Scottish Government as part of the LEO linkage – meaning the subsequent data extract shared through ONS is not linkable to other datasets out with LEO.

Linkage Quality and Limitations

The matching exercises between the education and DWP/HMRC datasets typically achieve a very high match rate. These vary between data sources and cohorts but are c. 95% matched on average. Almost all of these matched individuals have at least one record in the employment or benefits datasets.

The vast majority of matches are accurate. However, due to inherent data quality and coverage issues of the large, matched administrative datasets utilised within the LEO process (caveats outlined in this guide), a very small proportion of incorrect matches may have been made. Some genuine matches will also have been missed.

The matching algorithm itself relies on multiple fields being accurately populated across both datasets. The match rates will differ from one education source to another and vary across certain learner characteristics as well as across the full academic history of data available. One reason for this is that CIS holds the most recent names and addresses for individuals. As

these details are updated, there will be a lower match rate for some of the historical data records matched within this dataset.

Learner characteristics that impact the match rate are: ethnicity, location and first language. 97% of white learners found a match in CIS compared to around 92% of non-white learners across all education sources. Chinese learners typically have a lower match rate of around 80%, this is particularly apparent in the HESA data. Learners from London have a lower match rate at 93% compared to all other regions at 97%. The employment and earnings data has been derived from administrative IT systems which, as with any large-scale recording system, are subject to possible errors with data entry and processing. The core purpose of this process is to collect tax from those who are eligible to pay it through this mechanism, which results in incomplete coverage of earnings data. Employers were previously not required to supply information to HMRC for individuals who earn below the tax threshold, although for large employers, these individuals are thought to be included due to common methods of data transfer to HMRC.

The benefits data included is taken from DWP administration systems. Due to the size and technical complexity of these systems, cuts of the data are taken at regular intervals. These are used to create longitudinal datasets, with start and end dates of benefit claims and the type of benefit.

Geography region data is accurate from 2012/13 when the current DWP system went live (October 2012). It is possible that address information is accurate before 2012/13, but only if individuals did not move from this address.

All data used in this process is drawn from administrative sources and consequently takes time to process and collate.

5. Research Potential

The **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** provides a powerful resource for analysing how higher education experiences shape graduates' employment, earnings and wider life-course outcomes. By linking Higher Education Statistics Agency (HESA) records with HMRC and DWP data, the dataset enables robust longitudinal analysis of graduate destinations, labour market transitions and participation in employment or benefits systems.

Although the Scottish version of LEO is at an earlier stage of development than its English counterpart, it shares the same underlying structure and therefore supports many of the analyses conducted in England. These include examining educational pathways, internal migration, early-career earnings, inequalities in outcomes and regional or institutional variations in labour market trajectories. As the Scottish dataset expands – particularly with the addition of school-level data – its analytical potential will continue to grow.

For Scotland, the dataset can directly inform priority policy areas such as [Getting It Right For Every Child \(GIRFEC\)](#), [The Promise](#) and [Bringing Hope, Building Futures](#) and the [Scottish Attainment Challenge](#). It can also contribute to future Scottish Government [Areas of Research Interest](#) (ARIs), including those planned around The Promise (focused on care-experienced young people) and advanced learning and skills analysis. Further, it aligns with ADR Scotland's wider Strategic Research Themes, particularly *Children and Young People* and *Work, Welfare and Economy*, and contributes to cross-cutting themes related to poverty and equalities.

What Research is Possible

The dataset supports a wide range of analytical questions across education, labour markets, inequalities and economic outcomes. Its key strengths include:

Longitudinal analysis

The multi-year linkage enables researchers to follow individuals over time, examining transitions from university into employment, sustained participation in the labour market and changes in earnings trajectories. Longitudinal data allow for both theoretical and policy-focused research, including life-course analysis.

Large, population-level coverage

The extensive sample size supports detailed analysis of specific sub-groups and geographies, such as rural areas, care-experienced students, certain degree pathways or marginalized groups. This enables more granular and equity-focused research.

Natural experiments and comparative analyses

The dataset can support quasi-experimental designs, including:

- Comparisons before and after national policy interventions
- Contrasts between graduates from different local authorities
- Analysis of how institutional, subject or regional differences shape longer-term outcomes.

Pathways, progress and participation

The data allow examination of progression through university, dropout patterns, postgraduate participation and articulation routes. Linking these pathways to labour market data creates opportunities to understand the full continuum from higher education through early and mid-career outcomes.

Labour performance and occupational mobility

The inclusion of employment, earnings and Standard Industrial Classification (SIC) codes enables exploration of occupational types, professional progression, migration for work and the relationship between degree subjects and longer-term outcomes.

These capabilities make the dataset highly relevant for Scottish policy priorities relating to skills, productivity, widening access, social mobility, regional development and economic participation.

Example Research Questions

Below are illustrative research questions aligned with Scotland's policy context and the structure of the dataset:

Labour market outcomes and transitions

- How do earnings and employment outcomes differ for graduates by subject, institution type or student characteristics?
- What are the early-career employment trajectories of graduates who move between Scottish regions or between Scotland and the rest of the UK?
- What is the relationship between area-based deprivation, educational participation and labour market outcomes?

Inequalities, protected characteristics and care experience

- How do outcomes differ for care-experienced students and how might these findings support ambitions under *The Promise*?
- Are there persistent earnings or employment gaps for students from disadvantaged backgrounds, similar to those observed in England (e.g., free school meal recipients)?
- How do gender, disability or ethnicity intersect with educational pathways to influence long-term outcomes?
- How does residential deprivation influence choice of course, likelihood of degree completion, and progression into the work environment?

Education pathways and post-study destinations

- How do different higher education pathways (e.g., articulation routes, part-time or postgraduate study) influence long-term outcomes?
- What factors predict sustained employment or progression into higher-skilled occupations?

- How has the earning potential associated with specific subjects or degree types changed over time?
- To what extent do socio-demographic characteristics explain variation in degree completion, dropout patterns or transitions into further study?

Benefits, economic participation and longer-term outcomes

- How likely are graduates to receive labour market related benefits in the years following graduation, and how does this vary across groups?
- What are the long-term advantages of a university degree for older graduates compared with younger graduates?
- How do outcomes differ between graduates living in rural areas and those in urban centres?

Previous Use Cases

As a relatively new dataset, published Scottish analyses are limited. However, current work by researchers within ADR Scotland provides examples of emerging use:

- A project led by Edinburgh Napier University is examining transitions of nursing graduates into clinical employment, identifying the proportion entering health-related occupations, differences in early-career trajectories and factors associated with non-completion. These analyses will inform higher-education planning, workforce development, and the effectiveness of bursaries.

To understand the wider potential of LEO, researchers can also draw on established work using the England LEO dataset, including:

- Thomson, D. (2025). [*Getting started with LEO: studying the post-16 activities of young people*](#). ADR UK Blog.
- Department for Education (DfE). (2023). [*Exploring educational attainment, internal migration, and economic outcomes in English travel to work areas in 2018 to 2019*](#). Official Statistics, GOV.UK.

- Tierney, J., & Bathgate, H. (2022). [*Understanding earnings gaps for free school meals recipients in England: 2017 to 2019*](#). Office for National Statistics (ONS).

These examples illustrate the breadth of analyses that can be replicated or adapted in Scotland as the **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** matures and becomes more widely accessible.

6. Accessing the Data

How to Access the Data

The **Scottish Longitudinal Education Outcomes (LEO) – University extract dataset** can be accessed via the ONS SRS. Researchers will need to be UK Statistics Authority (UKSA) accredited under the Digital Economy Act 2017, or an approved researcher through the Statistics and Registration Service Act 2007. Details on how to make an application for this data can be viewed here: <https://ons.metadata.works/projects/application>

This dataset is made available to researchers as a large csv file, that allows the data to be worked with using a variety of analytical software such as R or Stata.

Although this dataset cannot be linked to other datasets, additional Scottish education data can be included in the LEO linkage process (see 4. Data Linkage Methodology) to further increase the scope of the Scottish LEO data, such as data pertaining to pupils at Scottish schools.

Additional datasets that have already been linked (Scottish LEO Colleges dataset and Scottish LEO Modern Apprenticeships dataset) are not yet available for research use. However, further work will be carried out in collaboration with the data owners of the Scottish college and modern apprenticeship data (The Scottish Funding Council (SFC) and Skills Development Scotland (SDS) respectively), to allow for this data to be used more widely by researchers.

It should be noted that in Scotland, approximately one third of higher education is delivered at Scottish Colleges, meaning the full outcomes of all higher education in Scotland is not currently captured.

Application and Approvals

Applications to access this dataset through the ONS Secure Research Service (SRS) must be submitted via the ONS Research Accreditation Service. ONS acts as the front-door for all LEO-related applications, coordinating the required checks and facilitating review by the

Scottish Government and other data owners. Once an application has been submitted, it progresses through the following three-stage approvals process:

Stage 1: ONS Feasibility and Standard Checks

ONS reviews the application to ensure it is complete, feasible and meets the accreditation standards required for access. This includes an ethics assessment by the Centre for Applied Data Ethics.

Stage 2: Confirmation of Correct Use of Data – SG, HMRC, DWP

Following the ONS feasibility checks, the Scottish Government undertakes an independent review and seeks approval from the other LEO data owners (HMRC and DWP) to confirm that the proposed use of their data is appropriate.

Stage 3: Approval of Research Aim – Research Accreditation Panel (RAP)

Applications that are endorsed by the data owners are then considered by the Research Accreditation Panel (RAP), which scrutinises the research aims and either accredits or rejects the application.

Output Clearance and Acknowledgements

All research outputs will be shared with Scottish Government for information purposes mainly. If there are any identified issues with research outputs, the Scottish Government will engage with the researcher to ensure data has been used appropriately.

Support and Contact Information

For support with access or accreditation, please refer to the ONS guidance on becoming an accredited researcher:

<https://www.ons.gov.uk/aboutus/whatwedo/statistics/requestingstatistics/becomeanaccreditedresearcher>

To discuss the Scottish Government review stage of the approval process, please contact:

FHEstatistics@gov.scot

7. Appendix

Supporting Documentation and Metadata

- **Metadata Catalogue Entry:** The dataset's metadata record is held on ONS Metadata Works (access requires login).
<https://ons.metadata.works/browser/dataset/1861122/0?tab=descriptiveMetadata>
- **Data Dictionary:** Available through ONS Metadata Works (access requires login).
<https://ons.metadata.works/browser/dataset/1861122/0/2091935>
- **Methodology Document:** The LEO methodology documents will be published by Scottish Government in Summer 2026 and will be accessible via:
<https://www.gov.scot/collections/colleges-universities-young-people-training-and-employment-statistics/>. If access to the methodology documentation is required before publication, please contact FHEstatistics@gov.scot.
- **Relevant National LEO Documentation (England / UK Context):** Describes the structure of LEO data and key outcome measures used across Further and Higher Education in England. <https://www.gov.uk/government/collections/statistics-outcome-based-success-measures>
- **Graduate Outcomes Survey Documentation:** Useful for comparison with LEO earnings / destination data. <https://www.hesa.ac.uk/innovation/outcomes> and <https://www.graduateoutcomes.ac.uk/>

Research Outputs and Insights

At the time of writing, Scottish specific LEO outputs are under development. However, several UK-level and policy relevant resources exist that illustrate the use of LEO data more broadly:

- **Data Insight (LEO – England):** Investigating the relationship between higher education participation, personal characteristics and area of residence among school or college

leavers: <https://www.adruk.org/news-publications/publications-reports/investigating-the-relationship-between-higher-education-participation-personal-characteristics-and-area-of-residence-among-school-or-college-leavers/>

- **ADR UK Research Fellows projects utilising Longitudinal Education Outcomes (LEO):** <https://www.adruk.org/our-work/browse-all-projects/adr-uk-research-fellows-longitudinal-education-outcomes/>
- **DfE LEO Research and Official Statistics:** Employment, earnings and destination analyses using LEO for England.
<https://www.gov.uk/government/collections/statistics-outcome-based-success-measures>
- **FE Week coverage of 2015 UK manifesto commitments on earnings and destinations** (contextual to the creation of LEO): <https://feweek.co.uk/utcs-apprenticeships-and-destination-data-central-to-tory-manifesto-plans-for-fe/>

Code and Technical Resources

No public code repositories specific to the Scottish LEO dataset were identified through search results. This is expected because:

- LEO data can only be analysed within Trusted Research Environments (TREs).
- Code written inside TREs cannot usually be published directly unless disclosure-checked.

However, relevant technical guidance includes:

- **Graduate Outcomes technical documentation** (for comparison with survey-based outcomes): <https://www.hesa.ac.uk/innovation/outcomes>
- **DfE methodological and statistical documentation for LEO-based outputs** (includes pseudonymisation rules, linkage and cleaning approaches):
<https://www.gov.uk/government/collections/statistics-outcome-based-success-measures>

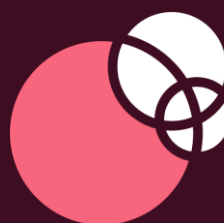
Contact and Upload Channels

- **Primary contact for the dataset** (metadata and technical queries):
FHEstatistics@gov.scot
- **ADR Scotland Engagement & Impact Team** (policy engagement and communities):
ADRScotland@gov.scot
- **ONS Metadata Works** (dataset metadata corrections or additions, general access and approvals): SRS.customer.support@ons.gov.uk

About ADR Scotland

ADR Scotland is transforming how public sector data in Scotland is curated, accessed and explored, so it can deliver its full potential for policymakers and the public. Combining specialists in the Scottish Government with the Scottish Centre for Administrative Data Research (SCADR), together they are making administrative datasets more readily linkable and conducting public good research safely and securely on a suite of critical issues in Scotland.

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